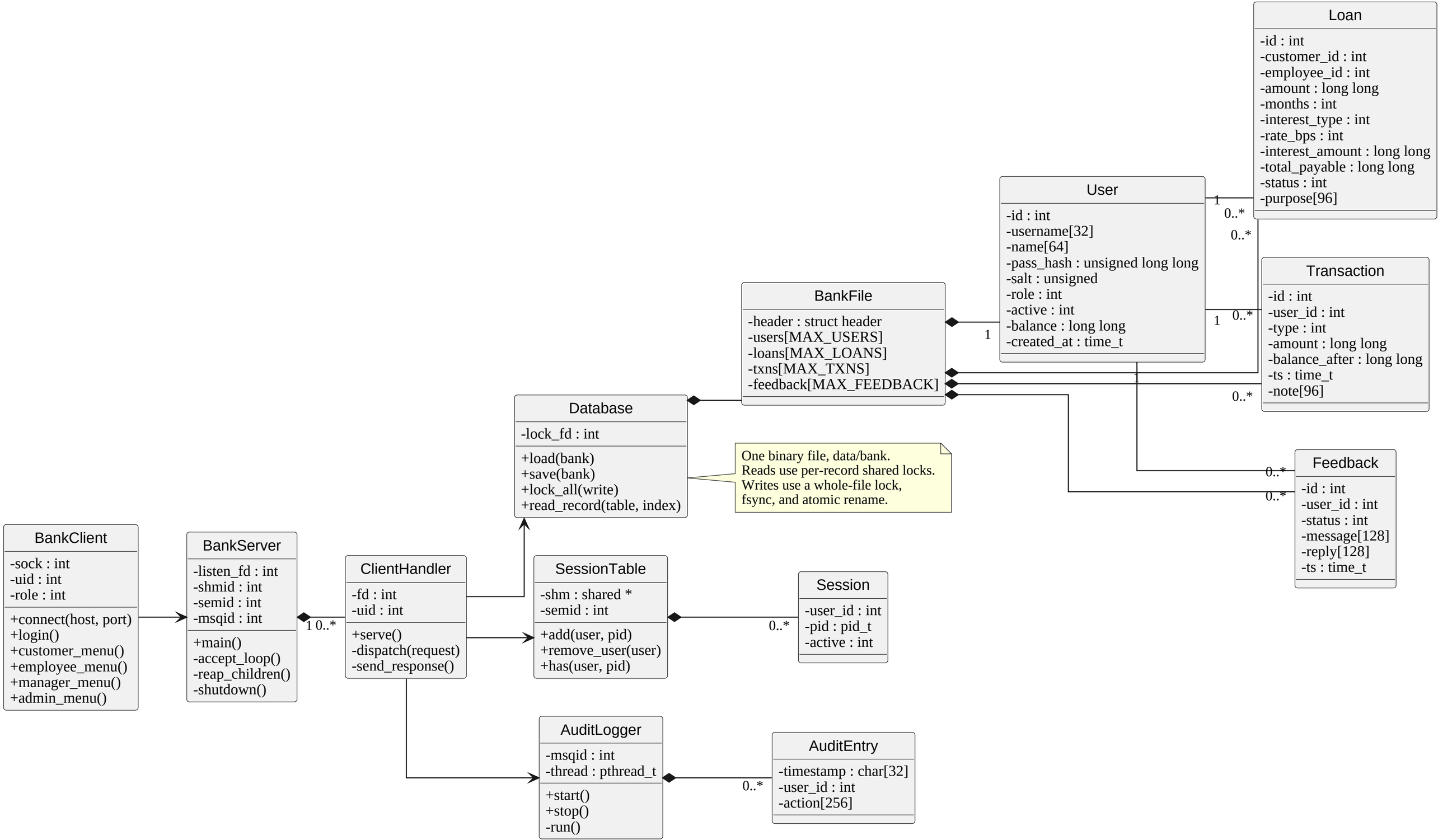
Loan
-id : int
-customer_id : int
-employee_id : int
-amount : long long
-months : int
-interest_type : int
-rate_bps : int
-interest_amount : long long
-total_payable : long long
-status : int
-purpose[96]
User
-id : int
-username[32]
-name[64]
-pass_hash : unsigned long long
-salt : unsigned
-role : int
-active : int
-balance : long long
-created_at : time_t
Transaction
-id : int
-user_id : int
-type : int
-amount : long long
-balance_after : long long
-ts : time_t
-note[96]
BankFile
-header : struct header
-users[MAX_USERS]
-loans[MAX_LOANS]
-txns[MAX_TXNS]
-feedback[MAX_FEEDBACK]
Database
-lock_fd : int
+load(bank)
+save(bank)
+lock_all(write)
+read_record(table, index)
One binary file, data/bank.
Reads use per-record shared locks.
Writes use a whole-file lock,
fsync, and atomic rename.
Feedback
-id : int
-user_id : int
-status : int
-message[128]
-reply[128]
-ts : time_t
BankClient
-sock : int
-uid : int
-role : int
+connect(host, port)
+login()
+customer_menu()
+employee_menu()
+manager_menu()
+admin_menu()
BankServer
-listen_fd : int
-shmid : int
-semid : int
-msqid : int
+main()
-accept_loop()
-reap_children()
-shutdown()
ClientHandler
-fd : int
-uid : int
+serve()
-dispatch(request)
-send_response()
SessionTable
-shm : shared *
-semid : int
+add(user, pid)
+remove_user(user)
+has(user, pid)
Session
-user_id : int
-pid : pid_t
-active : int
AuditLogger
-msqid : int
-thread : pthread_t
+start()
+stop()
-run()
AuditEntry
-timestamp : char[32]
-user_id : int
-action[256]
1
0..*
1
0..*
1
0..*
1
0..*
1
0..*
1
0..*
1
0..*
1
0..*